



Predictive Analytics: Project

The University of Texas at Dallas

Naveen Jindal School of Management

MKT/BUAN 6337.003

Submitted on: 05/05/2022

Submitted to: Dr. B.P. Murthi

Submitted by:

Group 8:

Akshai Seshadri

Ayush Chandel

Kailyn Dang

Rachel Denman

Manvi Mehta

Sakshi Nilesh Shah

I – Introduction & Data Pre-processing

Four data sets were provided with different variables and observations relating to the sale of coffee products in different retail markets. The coffee_grocery dataset had 5,938,038 observations and 11 variables; key variables in this dataset include: units (total unit sale), dollars (total dollar sales), feature, display, and price reduction. The delivery_stores dataset had 2051 observations and 7 variables with key variables regarding variations of the brand, package, and form. The prod_coffee dataset had 6929 observations and 21 variables detailing market (location), and grocery store chain. The IRI_week_translation dataset had 526 observations and 6 variables connecting data to the exact week and month of sale.

We initially evaluated the **summary statistics** of each dataset to comprehend the data presented before performing Data Preprocessing to avoid bias in the models we ran. We discovered that the data sets 'prod coffee' and 'coffee groc' could be **combined using UPC codes**. We tested for **missing values** after combining the aforementioned dataset using UPC codes and discovered that prod coffee had 3682 missing values. We eliminated them from the dataset because the number of missing values was less than 30% of the total number of observations.

Removing Outliers

To check for outliers we used the percentile method. The gap between the 99th percentile and 100th was large for VOL_EQ, DOLLARS, UNITS and price. In addition to the highest one percentile, we checked the lowest one percentile. Of the four variables, only two had large gaps in the bottom one percentile, Dollars and price.

Column	0 Percentile	1 Percentile	99 Percentile	100 Percentile
VOL_EQ	.0438	.0875	2.4375	8.625
DOLLARS	.02	1.24	185.69	7900.81
UNITS	1	1	53	4013
price	.01	.99	9.99	44.95

Before removing the outliers there were 5937811 observations. After removing the outliers, there are 5581164 observations.

II – Descriptive Statistics and Project Scope

Objectives:

- Industry analysis on factors driving sales of coffee (testing various marketing promotions)
- Comparison of leading coffee brands and analyzing factors driving sales for each
- Studying the customer preference for a particular brand and analyzing factors driving the purchase decision

For this project, we are representing **Starbucks**. Starbucks has \$12,280,943 in sales with 1,605,716 units sold. Starbucks has a **market share of 9.07%, ranking 3rd** amongst all brands in the category (coffee). The coffee category has a total of \$140,528,292 in sales across 50 markets and 130 grocery and drugstore chains. The average price across all coffee products is \$4.69, with Starbucks being on the higher end of the pricing scale with an average of \$8 and average size of 12oz compared to the standard 14oz.

Brand	Units	Avg. Price	Avg. Size	Dollars
MAXWELL HOUSE	5,843,729	4	17	22,218,100
FOLGERS	3,809,170	4	20	15,230,883
STARBUCKS	1,605,716	8	12	12,280,943
PRIVATE LABEL	3,167,609	4	15	10,779,011
EIGHT O CLOCK	1,495,411	5	18	6,154,868

III - Models and Insights

Model 1- Effectiveness of Marketing Promotions

The goal of this model is to assess and evaluate the marketing efforts (price, promotions, and advertisement) of the coffee industry. This will help frame how these factors affect the industry and we will later evaluate how to effectively establish a differentiated marketing position within the industry as a whole.

Variable	Parameter Estimate	t Value	STB	Rank based on STB
Intercept	23.99475	1187	0	
Price per oz	-13.66408	-288.34	-0.1693	1

Price sq	2.26856	103.44	0.05973	6
Price Reduction	0.53181	19.45	0.00892	9
Large size ad	16.79435	239.48	0.10294	3
Retailer coupon or rebate	18.7484	109.75	0.04501	7
Medium size ad	11.28394	170.54	0.07342	5
Small ad	8.68116	52.22	0.02148	8
Minor Display	19.79829	187.09	0.07685	4
Major Display	28.53224	330.5	0.13693	2

The first regression models the relationship between Sales, and all the independent variables as follows:

$$\text{Sales} = \beta_0 + \beta_1 \cdot \text{Price} + \beta_2 \cdot \text{PriceSq} + \beta_3 \cdot \text{PR} + \beta_4 \cdot \text{Fa} + \beta_5 \cdot \text{PaPlus} + \beta_6 \cdot \text{Fb} + \beta_7 \cdot \text{Fc} + \beta_8 \cdot \text{D1} + \beta_9 \cdot \text{D2}$$

● **Coefficient interpretation:**

- The model intercept is 23.10 meaning that if all the independent variable inputs were 0, sales would be \$23.10.
- When there is an increase of \$1 in price, the sales reduces by \$13.66- **Price per oz is the most important contributing factor according to Standardized betas in the coffee industry.** We use the squared price to capture the nonlinear effect of price on Sales. As the price vs Sales plot would be an inverted U-shaped curve. In the coffee industry, the sales will decrease when price is increases and then the sales increases even though the price increases. When there is an increase of \$1 in price, the sales reduces by \$13.66.
- Display 2- Major (D2) has the **2nd most important** contributing factor to sale, and the t Value is significant. When any store chooses to do in-store display for their Brand/ product (Display major), sales increases by \$28.50 as opposed to not having any display.
- When a store features a brand in flier with Large size advertisement, results in a \$16.80 increase in customer spend or sales. This is the **3rd most important** contributing factor to sales.
- When any store chooses to instore display for their Brand/ product (Display minor), sales increases by \$19.80 as opposed to not featuring an ad on the flier. This is the **4th most important** contributing factor to sales.
- When a store features a brand in flier with Medium size advertisement, results in a \$11.30 increase in customer spend or sales. This is the **5th most important** contributing factor to sales.

- When a store features a brand in flier with A+ size advertisement – also known as “Q” or “R” – retailer coupon or rebate and small ad (usually 1 line of text), results in a \$18.7 and \$8.7 increase in customer spend or sales respectively. This is the **7th and 8th most important** contributing factor to sales.
- Price reduction or discounts has only a \$0.531 increase on sales, being the **9th most important factor** and is significant as well.

Model 2- Brand Analysis

Running a T-TEST using Dollars variable to ensure that average sales of Maxwell and Starbucks are significantly different. The output shown below has a p-value less than 0.05 for equality of variance, indicating that the Satterthwaite technique must be used to determine the significant difference of unequal variance. The p-value for it is also less 0.05, indicating that the sales of Starbucks and Maxwell are substantially different.

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	1.40E+06	-113.72	<.0001
Satterthwaite	Unequal	779175	-110.32	<.0001
Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	434138	965259	1.17	<.0001

	STARBUCKS	MAXWELL	FOLGERS
Attribute	Parameter Estimates	Parameter Estimates2	Parameter Estimates3
Price per oz	2.02	-8.35	-110.01
Price reduction	2.79	2.01	3.31
Large size ad	14.39	14.84	9.55
Retailer coupon or rebate	7.76	8.80	32.79

Medium size ad	7.46	14.50	9.06
Small ad	3.18	5.76	15.17
Minor Display	19.10	23.16	36.34
Major Display	18.84	30.12	38.46
Coffee type- Ground	8.48	0.20	-3.04
Foil bag packaging	7.62	-4.90	-
Flavor/Scent House Blend	4.06	-1.78	-
Flavor/Scent French Roast	6.16	-2.35	-
Flavor/Scent Breakfast Blend	5.20	-	-

For our brand study, we are comparing Starbucks to Maxwell and Folgers because they are competitors to Starbucks and Maxwell and Folgers rank first and second in total sales and total units, respectively. Analyzing what works well for the top two companies in the business will offer an overview of what works better in the coffee industry for Starbucks.

- According to the statistics above, **Starbucks has price elasticity lesser than one**, indicating that if the business decides to raise its price, there is a chance that its sales will increase **assuming that there is no unobserved variable that has an influence on pricing**. Unobserved factors, such as **brand loyalty**, may exist. If Starbucks increases prices, sales may increase because the fall in quantity demanded is much lesser than the increase in price. This means that it has customers who are not price sensitive and are particularly loyal to Starbucks.
- Given that Folger's has the second highest sales in the coffee industry, Folgers appears to stimulate sales by offering products at a lower price. Folger's price has a greater influence on sales than Maxwell and Starbucks when the price per oz is increased by \$1. For Folgers, a one-dollar price increase reduces sales by 110.01 dollars. This indicates high price elasticity.
- Currently, a unit rise in Starbucks retailer coupon may raise sales by 7.76 dollars; nevertheless, it can be noted that a unit increase in Folgers retailer coupon has an impact that is more than three times that of Starbucks.
- Starbucks should focus more on Major Displays for promotions since one unit increase in Major Display may improve Starbucks sales by \$18.84. This should be an industry-specific relationship,

as for Maxwell and Folgers, Major Display increases sales more than no display. When we look at minor displays, the influence on the result is not as large as major displays when compared to no displays, but it is still substantial.

- Foil bag packaging has adverse effects on sales for Maxwell but positive for Starbucks. This essentially shows that what works for one company in the industry may not work for another given their different brand positioning and packaging.
- The flavors house blend, french roast and scent breakfast blend have a positive impact on dollars as compared to other flavors offered by Starbucks. However, the same is not true for Maxwell.

Model 3- Starbucks Decision Model

It's important for Starbucks to know the driving factors which induce the decision to purchase their products. While customer data was not available to construct a multinomial brand decision model, a model was constructed to determine some factors that induced the choice of Starbucks compared to all other brands. Because Starbucks only makes up around 9% of the market sales, in order to run a logistic regression model, modeling Starbucks choice (1 if Starbucks was chosen, 0 otherwise), oversampling was performed. The oversampling created a 50/50 split between Starbucks decision and otherwise. The model had a percent concordance of 87.1% and the McFadden R squared is 42.54% indicating that the model fit is good.

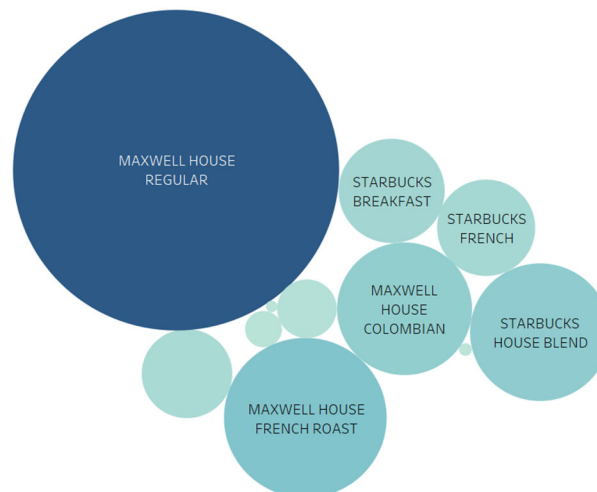
Odds Ratio Estimates					
	Point Estimate	Pr > ChiSq		Point Estimate	Pr > ChiSq
Price reduction	0.542	<.0001	Bean	<0.001	0.9978
Large size ad	1.956	<.0001	Extra	<0.001	0.9851
Retailer coupon or rebate	1.691	<.0001	Fine G	<0.001	0.93
Medium size ad	2.758	<.0001	Ground	1.897	<.0001
Small ad	1.76	<.0001	Bag packaging	0.787	0.0061
Minor Display	3.074	<.0001	Can	<0.001	0.6463

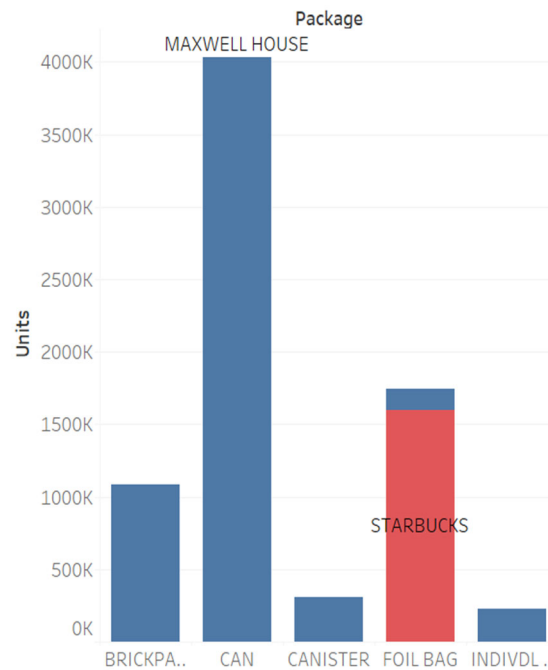
Major Display	0.655	<.0001	Individual Wrap	<0.001	0.9181
			Resealable Bag	0.078	<.0001

Using the odds ratio, we can interpret how each variable affects the odds of Starbucks getting chosen over other brands. Statistically significant variables are interpreted as follows:

- A price reduction that week will decrease the odds of Starbucks being chosen over brands by 45.8% compared to no price reduction
- A large sized ad (Feature A) will increase the odds of Starbucks being chosen by 95.6% compared to having no features
- A retailer coupon or rebate will increase the odds of Starbucks being chosen by 69.1% compared to having no feature
- A medium sized ad (Feature B) will increase the odds of Starbucks being chosen by 175.8% compared to having no feature
- A small sized ad (Feature C) will increase the odds of Starbucks being chosen by 76% compared to having no feature
- A minor display will increase the odds of Starbucks being chosen by 207.4% compared to having no display
- A major display will decrease the odds of Starbucks being chosen by 34.5% compared to having no display
- The coffee being sold in “ground” form (f4) will increase the odds of Starbucks being chosen by 88.7% compared to other forms
- The package being in a bag will decrease the odds of Starbucks being chosen by 21.3% compared to other packages
- The package being in a reseal bag will decrease the odds of Starbucks being chosen by 92.2% compared to other packages

IV – Conclusions & Recommendations



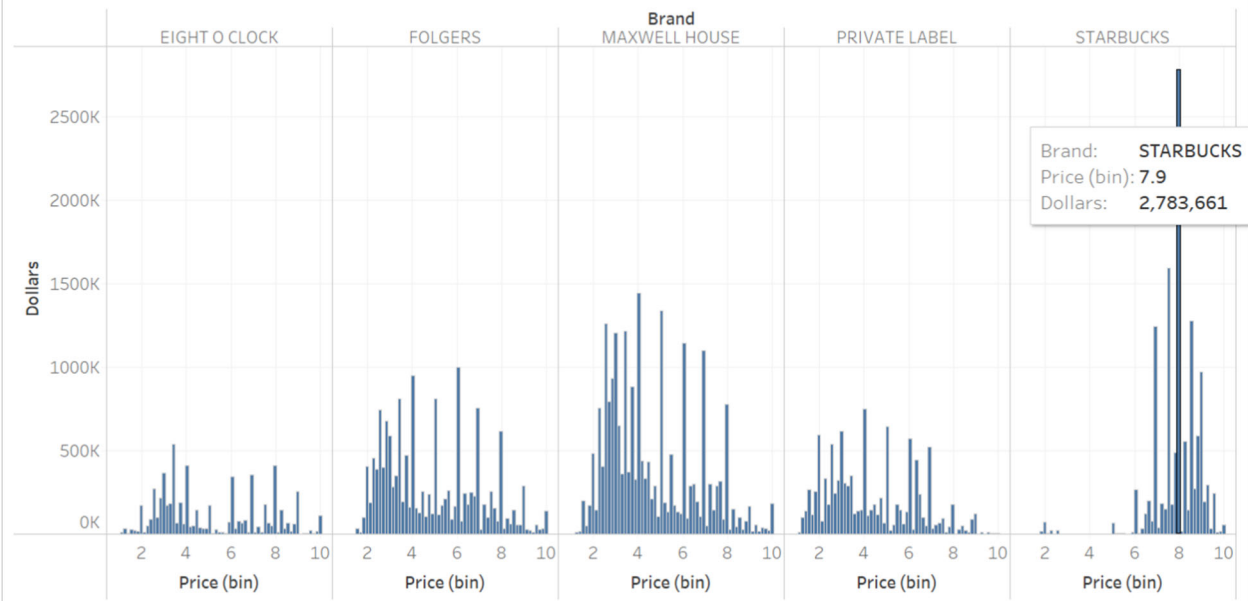


The visual inspection reveals Brand Maxwell House's Regular flavor is the most frequently ordered coffee in the market. The popularity of coffee orders is as follows, Maxwell House's French Roast coffee closely followed by Starbucks House Blend. Maxwell House's Columbian sold a similar number of units to Starbucks house Blend. Starbucks second two most popular flavors were Breakfast and French coffee which had very similar consumption amounts. While Maxwell House Regular was clearly the most popular flavor, Maxwell House and Starbucks actually follow similar patterns. Of the six flavors listed, both brands had their original flavor and French Roast flavor made the cut.

While studying the preferred packaging, these were the top five types of packaging. Starbucks exclusively used foil bags and Maxwell House used only a small amount of foil bags. Starbucks Foil Bag was the closest competitor to Maxwell House Can although the Can was preferred more than twice to Starbucks's Foil Bag. Brickpack was the next closely preferred option for packaging.

Maxwell House had a much larger overall sales than Starbucks so the flavors and packaging they used was going to be more popular than Starbucks. The interesting part is seen by the similarities and differences between the two rather than the amount. The chosen flavors are very similar between the two brands, but the packaging chosen by each brand is quite different.

Sheet 11



A Brand-Price analysis reveals that Starbucks sales revenue is highest when priced at \$8, however, as suggested by the linear regression model, it follows an inverted U-shaped curve. Folgers and Maxwell House customers are on the other hand price sensitive as they both follow symmetrical patterns. As the price increases beyond a certain point, \$6 and \$4 respectively, the revenue drops. This is in line with the analysis from models 2 and 3. Starbucks has effectively positioned itself as a higher end coffee brand, cultivating non-price sensitive customers.

Recommendations

- Though Maxwell and Folger's are leaders in terms of \$ sales in the industry, Starbucks has its unique proposition which can be leveraged to increase sales in the future. Starbucks has a loyal, high income consumer base that perceives coffee as an affordable luxury. A small price increase (\$1) can have a positive impact(\$2.02) on their revenue without decreasing demand. However, it does need to focus on its marketing strategies as its competitor Maxwell can leverage that more easily to increase sales (based on the coefficients of the regression model). Starbucks can gain market share (and sales) in 2 ways:
 - 1. Introduce a cheaper product to compete with Maxwell and Folgers.** By introducing a cheaper product line, Starbucks will closely compete with the market leaders based on price and customer segments, but must be led with caution as it may lead to cannibalization and dilute the "luxury" brand image that Starbucks has developed.

2. **Focus on their current customers.** Starbucks can gain more sales by focusing on creating new products to fit its current customer needs. Because they have a less price sensitive and loyal customer base, Starbucks can retain and increase market share by focusing on customer retention and increasing the frequency and volume of which they buy.

- In terms of marketing and promotions, a price reduction and large sized ad features both decreased the odds of Starbucks being chosen over other brands. Minor display ads and mediums features were the most persuasive methods. This furthers the idea that Starbucks customers are composed of the least price sensitive customer segments, opting to purchase higher priced coffee products with less promotional efforts. Further analysis on messaging and the content of these features and ads to better target its customers may be advantageous to explore.